

Sets and Logic

Previous Lecture

- *Venn diagrams*

- *Operations of*

- *Intersection*
- *Union*
- *Symmetric difference*
- *Complement*
- *Difference*

- *De Morgan's law for sets* $\overline{A \cup B} = \bar{A} \cap \bar{B}, \quad \overline{A \cap B} = \bar{A} \cup \bar{B}$

Predicates and their meaning

- Consider the predicates

$$P(x,y) = x > y,$$

$$Q(x,y) = y < x$$

Are they different?

- Extensional** view: a predicate is defined over a given domain by the set of values **on which it is true** (called its **models**)

$$P(x) \equiv \{ x \mid P(x) \} = P_x$$

“equivalent”

- Now we can formally make a connection between logic and set operations

$$P(x) \wedge Q(x) \equiv \{ x \mid P(x) \wedge Q(x) \} = P_x \cap Q_x$$

Sets and Logic

● Operation correspondence

$$\neg P(x) \equiv \{x \mid \neg P(x)\} = \overline{P_x}$$

$$P(x) \vee Q(x) \equiv \{x \mid P(x) \vee Q(x)\} = P_x \cup Q_x$$

$$P(x) \oplus Q(x) \equiv \{x \mid P(x) \oplus Q(x)\} = P_x \Delta Q_x$$

$$0 \equiv$$

$$1 \equiv$$

● Law correspondence

$$P(x) \Leftrightarrow Q(x) \equiv P_x = Q_x$$

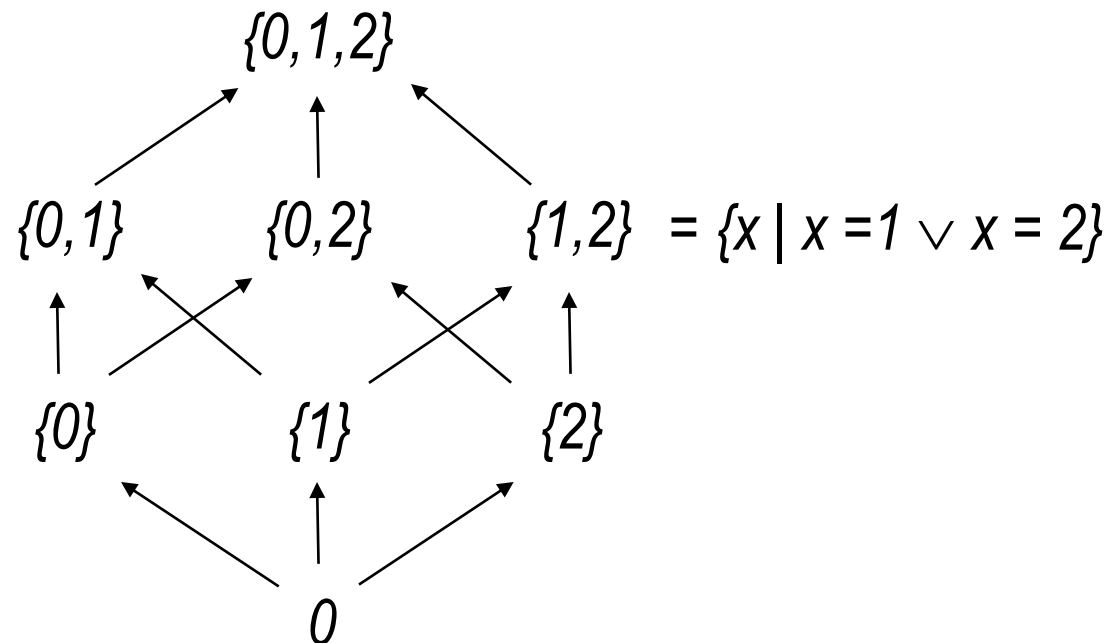
$$\neg(P(x) \vee Q(x)) \Leftrightarrow \neg P(x) \wedge \neg Q(x) \equiv \overline{P_x \cup Q_x} = \overline{P_x} \cap \overline{Q_x}$$

Power Set

- Given a set A , the **power set** of A is the set of all subsets of A

$$P(A) = \{ B \mid B \subseteq A \}$$

- Example. Find $P(\{0,1,2\})$



Propositional models

- Given a set A of propositional variables, what might

A subset of A correspond to (in logic)?

Valuation (set of true propositions)

The powerset $P(A)$ correspond to?

Set of all valuations

A proposition p correspond to (in sets)?

Set of valuations in which p is true

A formula $p \wedge q$ correspond to?

Set of valuations in which $p \wedge q$ is true

- Example: $A = \{p, q\}$

	p	q	$p \wedge q$
$\{\}$	0	0	0
$\{q\}$	0	1	0
$\{p\}$	1	0	0
$\{p, q\}$	1	1	1

Rows = subsets of propositions

Cols = models of the formula

Cardinality of the power set

- *How many elements are in the powerset of a finite set?
==> how many lines in a truth table over n variables?*

● **Theorem.** *If A is a finite set, then $|P(A)| = 2^{|A|}$*

Proof (sketch)

Let B be a subset of A and $|A|=n$.

We can describe B by which of the n elements of A it contains!

Let $i_1i_2i_3...i_n$ be the binary string where $i_j = 1$ if and only if B contains the j th element of A

How many unique strings (and hence subsets B) are there?

- *Representation of a subset of A where $|A|=n$ as a binary string of length n is called an **indicator vector** or **characteristic function***

Example

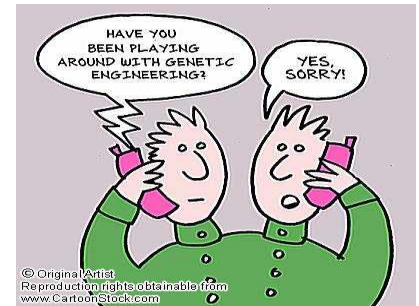
- Let $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
- Write the following as indicator vectors $i_1 i_2 i_3 i_4 i_5 i_6 i_7 i_8 i_9 i_{10}$
 - $\{1, 2, 3\}$
 - $\{1, 3, 5, 7, 9\}$
 - The even digits
 - $\{\}$
 - $\{1, 10\} \cup \{5\}$
 - $\{1, 2\} \cap \{1, 3\}$
 - $\overline{\{5\}}$

Relations

- The predicate $P(x,y) = x > y$ isn't a set of values in the same sense... it's a *relationship* that holds between the values of x and y

- Between people, familial relationships

`to be brothers'	x is a brother of y
`to be older'	x is older than y
`to be parents'	x and y are parents of z



- Between things, numerical relationships

`to be greater than'	$x < y$ on the set of real numbers
`to be divisible by'	x is divisible by y on the set of integers
- Between things and people, legal relationships

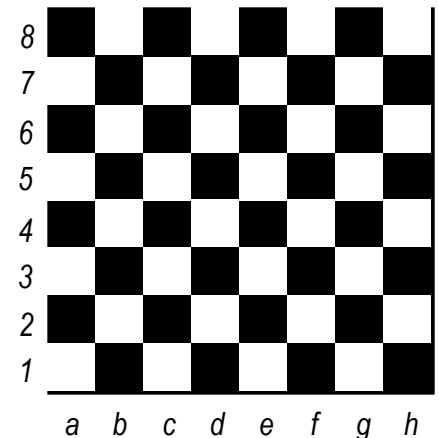
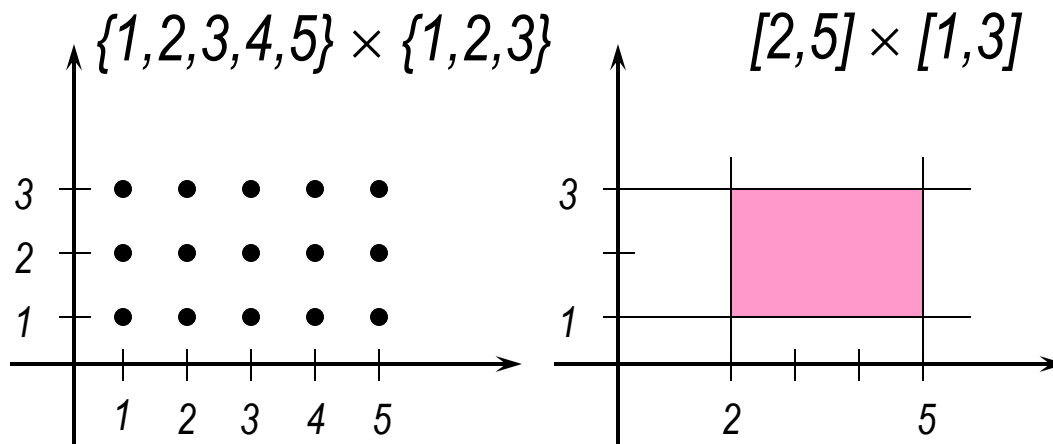
`to be an owner'	x is an owner of y
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Relations as sets: the Cartesian Product

- The **Cartesian product** of sets A and B , denoted by $A \times B$, is the set of all **ordered pairs** (a,b) of elements from A and B .

$$A \times B = \{ (a,b) \mid a \in A, b \in B \}$$

- The elements of the Cartesian product are ordered pairs. In particular, $(a,b) = (c,d)$ if and only if $a = c$ and $b = d$.
- If sets are thought of as '1-dimensional' objects, Cartesian products are 2-dimensional



Example

● *Are the following sets equal or not equal?*

$\{1,2,3\}$ and $\{3,2,1\}$?

$\{\{1,2,3\},\{3,2,1\}\}$ and $\{\{1,2,3\}\}$?

$\{(1,2,3)\}$ and $\{(3,2,1)\}$?

$\{(1,2,3), (3,2,1)\}$ and $\{(1,2,3)\}$?

Cardinality of Cartesian Product

● Theorem.

$$|A \times B| = |A| \cdot |B|$$

● *Proof*

Consider an element a of A

We can construct $|B|$ unique pairs (a,b) corresponding to the $|B|$ elements of B

Since we have $|A|$ such unique elements a , in total there are $|A| \cdot |B|$ unique ordered pairs.

Q.E.D.

Back to predicates and their meaning

- Consider the predicates

$$P(x,y) = x > y,$$

$$Q(x,y) = y > x$$

- They have 2 indeterminates, so they aren't quite sets of values in the same sense.
- They give a *relationship* between two indeterminates x and y , not just a set of values.
- Their models are hence *sets of ordered pairs*, i.e. a subset of the Cartesian product of the variable's domains called a *relation*

$$P(x,y) \equiv \{ (x,y) \mid P(x,y) \} \subseteq U \times U$$

Binary Relations

- A **binary relation** from set A to set B is any subset of $A \times B$.
If $A = B$ then we say that the relation is **on** the set A

' x is a brother of y ' $\subseteq$ $\text{People} \times \text{People}$

' x is older than y ' $\subseteq$ $\text{People} \times \text{People}$

' x is an owner of y ' $\subseteq$ $\text{People} \times \text{Properties}$

' $x < y$ ' $\subseteq$ $\mathbb{R} \times \mathbb{R}$

' x divides y ' $\subseteq$ $\mathbb{Z} \times \mathbb{Z}$



- **Binary predicates** $P(x,y)$ correspond to binary relations!

$$P(x,y) \equiv \{ (x,y) \mid P(x,y) \} \subseteq U \times U$$

Sets, Relations, and Predicates

- *Now sets, relations and predicates are essentially the same object!*
- *Sets, logic, and relational algebra (next week) all pretty much the same too!*

Unary predicate

$$P(x)$$

Set (“unary relation”)

$$A = \{ x \mid P(x) \} \subseteq U$$

Binary predicate

$$P(x,y)$$

Binary relation

$$R = \{ (x,y) \mid P(x,y) \} \subseteq U \times U$$

...or if you want to go further, all logic is equivalent to algebra (algebraic logic) is equivalent to Topos theory (generalized set theory inside of category theory) is equivalent to algebraic topology (continuous mathematics) is equivalent to programming languages (type theory) is equivalent to...